

軟體測試與資安檢測

教師：王凡

授課時間：每週三 9：10—12：00

博理館 113

課程代號：EE 5170

識別碼：921 U9550

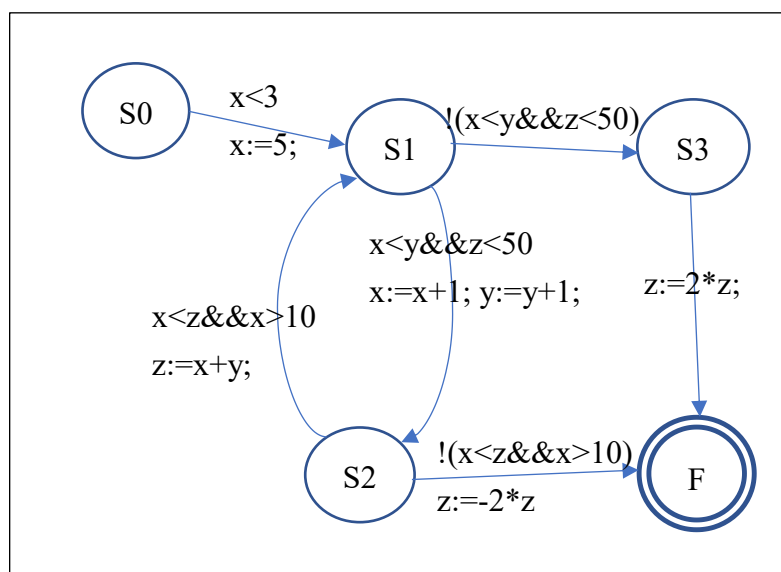
民國 110 年春季班

QUIZ (2021/04/28)

Open-book, open-internet, no discussion

Name: _____ Department : _____ Student ID: _____

Suppose we have the following sorting program.



Requirement 1

1. Considering the state-transition diagram in Requirement 1 in the above, please derive and write the condition on x, y, z at state S_0 using backward deduction for testing the triggering condition from S_3 to S_1 .

2. Continuing from problem 1, please write down the test input values to achieve RACC for the triggering condition from S3 to S1.

3. Considering the state-transition diagram in Requirement 1 in the above, please derive and write the condition on x, y, z at state S_0 using backward deduction for testing the triggering condition from S_3 to F .

4. Continuing from problem 3, please write down the test input values to achieve RACC for the triggering condition from S3 to F.

5. Consider the following predicate.

$$x < y \ \&\& \ (a \rightarrow x > 3) \ \&\& \ !(a \parallel (!a \ \&\& \ b \ \&\& \ \sim c))$$

Please write down the DNF of this formula. Then write down the variable valuations for implicant coverage.

6. Continuing from problem 5, please write down the variable valuations for UTPC.

7. Continuing from problem 5, please write down the variable valuations to achieve CUTPNFP.

8. Continuing from problem 5, please write down two valuations that can detect an instance of LOF fault class.

9. Continuing from problem 5, please write down two valuations that can detect an instance of LNF fault class.